

LEARNING OUTCOMES!

Candidates should be able to:

Internal transport

- interpret maps to describe the regional variations in the density and pattern of the road, rail and air transport networks within Pakistan
- explain the factors which help and hinder the location, maintenance and development of roads, railways and airports
- describe improvements that have recently taken place in road, rail and air communications, and consider the feasibility of new developments
- compare the advantages and disadvantages of road, rail and air transport within Pakistan for both goods and people
- evaluate the development of new transport schemes, including motorway and airport development.

International transport

- identify on a map those roads (including the name of the pass they use, where relevant) and railways which cross the international boundary and are in use for at least part of the year
- identify on a map the ports of Keamari, Qasim and Gwadar, and the cities with international airports
- explain the factors which affect the location and development of cross-border roads and railways, seaports, dry ports and airports
- explain what is meant by the term dry port, name an example of one and explain why they have been developed in many cities of Pakistan.

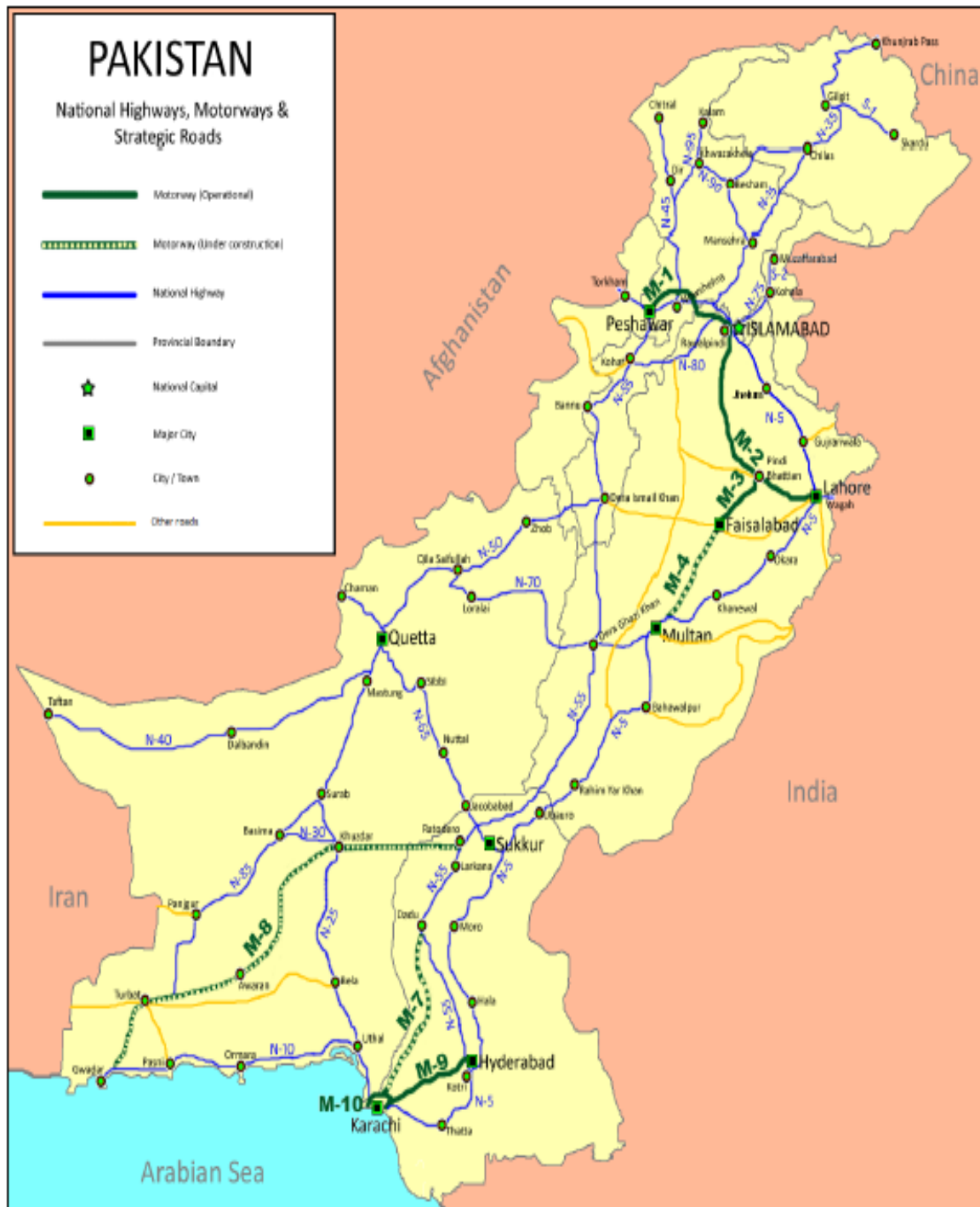
Telecommunications

- explain the importance of radio, television, phones, fax machines, emails and the internet in the fields of education, industry, services and trade
- understand the problems of providing telecommunications in some parts of Pakistan
- evaluate the role of telecommunications in the development of Pakistan.

ROAD NETWORK

- ✓ How roads are interconnected / joined
- ✓ linkage of roads / road pattern

ROAD NETWORK OF PAKISTAN



DISTRIBUTION AND DENSITY OF ROADS

Sindh

- ✓ In Sindh the major highways, N-55 (also known as Indus Highway) and N-5 run along both banks of Indus in the South-North direction.
- ✓ They connect Karachi, Thatta, Hyderabad and Sukkur etc
- ✓ There's a motorway that connects Karachi and Hyderabad. These two cities are two main foci of the road network in Sindh.
- ✓ M-10 is found near Karachi

Punjab

- ✓ As the highways in Sindh enter Southern Punjab they still follow both banks of Indus and then move away from Indus. As we move North, we see that many cities are interconnected with a web of dense road network along banks of rivers and also across the doabs, with main foci being Lahore, Faisalabad and Multan
- ✓ Two motorways connect Lahore with Faisalabad and with Islamabad.
- ✓ A major road also leads from Lahore to Wagah and eventually into India.
- ✓ Furthermore, two roads from Bahawalpur and Multan respectively, merge into one near the border and then lead to India

Baluchistan

- ✓ In Balochistan, the main focus is the city of Quetta, from which roads lead in all direction. To North we head to Chaman and Afghanistan. Another road leads West to Dalbandin and then to Iran.
- ✓ To North-East leads to Western Punjab and to Southern Khyber-Pakhtunkwa through Zhob. One road also leads to Sukkur in Sindh
- ✓ N-40 and N-25 form RCD highway connecting Quetta and Karachi with Iran
- ✓ Near the coast Makran Coastal highway connects Gwadar and other fishing centres like Pasni with Karachi

Khyber Pakhtunkhuwa

- ✓ The Indus Highway connects Dera Ismail Khan with Peshawar. From Peshawar a highway leads into Afghanistan at Torkham through the Khyber pass
- ✓ M-1 connects Peshawar with Islamabad, and then other highways connect Peshawar with Northern areas of Kalam and Chitral

Gilgit Baltistan and Azad Kashmir

- ✓ Karakoram Highway connects Hassan Abdal with China via Khunjerab Pass, after passing through Abbottabad and Gilgit
- ✓ Murree Express Highway connects Islamabad with Murree and Muzaffarabad

REASONS FOR THIS DISTRIBUTION

- ✓ The road network of Punjab is the densest followed by Sindh and then of Balochistan. This is because there is a lot of population in Punjab which needs to be connected.
- ✓ Balochistan has a rugged terrain along with a low population, which makes road building difficult and uneconomical (as they will be used by very few people)
- ✓ In Khyber-Pakhtunkhwa, the road network connects different valleys like of Peshawar with other areas. Like Balochistan, the rough terrain has limited the density of the road network
- ✓ There are more foci of road network in Punjab because many major cities are located here due to their economic prosperity
- ✓ The main pattern of road network in Sindh is South-North, because roads follow banks of River Indus and that there's only one river in Sindh. Also on East we have more or less an inhospitable area of the Thar desert

Why Few Roads in Mountains of Pakistan?

- ✓ Rugged Topography (making road building difficult)
- ✓ Steep slopes (making road building difficult)
- ✓ More expensive to build due to cuttings of mountains
- ✓ Unstable land - landslide / rock falls (dangerous)
- ✓ Low population density areas (so less demand for roads / / lack of labour to build roads)
- ✓ Groups who live in these areas may be traditional in outlook and therefore oppose areas being opened up by roads / are resistant to modernisation)
- ✓ Roads blocked by snow.

Problems of using road transport in the northern areas of Pakistan

- ✓ Mountain roads very narrow/small for large trucks
- ✓ Danger of falling rocks
- ✓ Likely to be blocked due to landslides
- ✓ Closed in winter due to snow
- ✓ Unmetalled roads/potholes

- ✓ Lack of security

Why Difficult to make new roads in deserted areas?

- ✓ Rugged Topography (making road building difficult)
- ✓ Steep slopes (making road building difficult)
- ✓ Low population density areas (so less demand for roads / / lack of labour to build roads)
- ✓ Extreme high temperatures making difficult working conditions for construction workers causing dehydration
- ✓ Lack of government investment/government funding;
- ✓ Opposition from tribal areas;
- ✓ Lack of security/insurgency;

IMPORTANCE OF ROAD NETWORK

- ✓ It is the most versatile form of network, meaning it can serve a lot of areas which rail and air systems can't serve, because laying out rail and air networks costs a lot of capital
- ✓ Provides door to door service
- ✓ Most of the goods transported inside Pakistan are on the road network, including commodities like oil from pipelines to city areas and to gas stations
- ✓ It can serve areas 24/7, there is no waiting for the rail carriage to arrive etc, goods can be transported any time
- ✓ It is cheap over short distances as compared to air or rail network
- ✓ It has little documentation involved which can be complex in rail and air transport
- ✓ There is little time wasted during uploading and receiving of goods, which is a common hassle in rail and air transport due to innumerable checks like for safety and for weighting the goods etc

LIMITATIONS OF ROAD NETWORK

- ✓ It is expensive and time consuming over long distances
- ✓ It can only carry goods in limited quantity as compared to air and especially rail transport
- ✓ It is also affected by traffic jams
- ✓ Like rail network it may not be able to serve some areas like the Northern Mountains which air transport can only serve. This is because of the harshness of the terrain, which makes it expensive to construct roads. Landslides are a common problem too.

RECENT DEVELOPMENTS AND FUTURE PLANS

- ✓ M4 is being constructed in Punjab; it will link Southern Punjab with the motorway network, by joining cities of Faisalabad and Multan. It will be completed in 2012, have 4 lanes and also will have the capacity to accommodate two further lanes
- ✓ M8 is being developed from Gwadar to Ratodero in Larkana Sindh. It will serve as a gateway to the Makran Coast. It will have 4 lanes
- ✓ Islamabad-Murree-Muzaffarabad Expressway is under construction. Only the Murree to Muzaffarabad section is yet to be built while the rest of expressway is done. It has 4 lanes
- ✓ Torkham-Peshawar Expressway is also planned. Torkham is the busiest port of entry between Pakistan and Afghanistan, it being a major shipping, transporting and receiving centre
- ✓ Sialkot-Lahore Expressway is also under consideration, improving trade between these two very important cities
- ✓ M-7 is being planned which will connect Rotadero with Karachi
- ✓ M-6 is being also planned. It will connect Dera Ghazi Khan with Rotadero, with Dera Ghazi Khan being an important centre for cotton textile and ghee industry
- ✓ M-5 will connect Multan with Dera Ghazi Khan, thus completing access of whole of Southern Punjab with the motorway network

Why was the first motorway in Pakistan built between Islamabad and Lahore?

- ✓ 2 large centres of population
- ✓ passed by other populated areas
- ✓ to promote growth of industrial estates
- ✓ large volume of traffic

Advantages Of Developing Motorways In Pakistan

- ✓ They help in promoting industrial growth as both products and raw materials can easily be transported between different cities. The operation of dry ports can also be improved because if any problems in the rail network occur; motorways can be used to transport the goods quickly to and from the port
- ✓ Industrial growth is encouraged in the areas which are near to the motorways. These motorways are wide (6 lanes) and have good quality surfaces, for smooth passage of traffic

- ✓ Congestion from other highway/roads is relieved and motorways might cut through a previous long route, which now becomes shorter. This helps industrialists in meeting orders on time.
- ✓ Tourists and other professionals (engineers) may find it easier to reach some areas. New tourist may bring in more money and employment for the local people
- ✓ Attract foreign investors

Disadvantages Of Developing Motorways In Pakistan

- ✓ Motorways are expensive to build and maintenance cost is high (as they are very long).
- ✓ Cost to economy.
- ✓ Furthermore, they require a lot of time to build.
- ✓ Industrial growth may be limited as motorways may not be the most suitable to for example a producer, who produces and sells his goods in bulk. He would prefer railways as roads cannot carry goods in bulk economically
- ✓ Also, these motorways don't always connect all major towns etc; they only connect major cities together. Thus the chronic poverty in rural areas may still not be solved

To what extent would the building of more motorways such as that between Lahore and Islamabad help the development of industry in Pakistan?

- ✓ For Possibilities - Advantages
- ✓ For Not Possible - Difficulties

It has been proposed that a new motorway should be built in Sindh from Hyderabad via Thatta to Keti Bandar on the Indus Delta, a town with a population of less than 25000.

Explain why this motorway might be needed. To what extent would this proposal be possible? [6]

Why needed

- ✓ New route to open up a seaport to relieve pressure on port of Karachi
- ✓ Enables quicker movement of local goods/people
- ✓ Stimulates industry (industrial estates along motorway / more efficient
- ✓ supply of raw materials/manufactured goods)

Possible

- ✓ Possible with high government/foreign funding
- ✓ Already Indus Highway as far as Thatta (and could more easily be upgraded to motorway)

Not Possible

- ✓ High cost
- ✓ Difficult to connect to coast/construct due to mangrove swamp (as would need deep foundations/high pillars/causeways)
- ✓ Prone to flooding
- ✓ Would involve felling mangrove forests (and destruction of vital habitat and a human resource)
- ✓ Ketī Bandar too small to be served by a motorway and therefore not economically worthwhile

To what extent could a better road network increase the development of small scale and craft industries of Pakistan? [6]

Advantages

- ✓ More remote areas can be connected e.g. Swat
- ✓ Good access to raw materials
- ✓ Greater access to training
- ✓ Greater access to (export) markets,
- ✓ Increases tourist market
- ✓ Faster / safer trade

Problems

- ✓ Limited marketing skills / access to product markets
- ✓ Roads likely to be often closed in mountain areas due to landslides / snow
- ✓ Cost of transport as middle man do not have own transport

Read the following article:

The Lowari Tunnel is due to open in 2017. It is 8.6km long and will give Chitral Valley its only all-weather road to the rest of Pakistan. For some in Chitral Valley this tunnel will be of real benefit but for others it will create problems.

To what extent does the Lowari Tunnel benefit or create problems for the local people and economy of Chitral? Give reasons for your answer. [6]

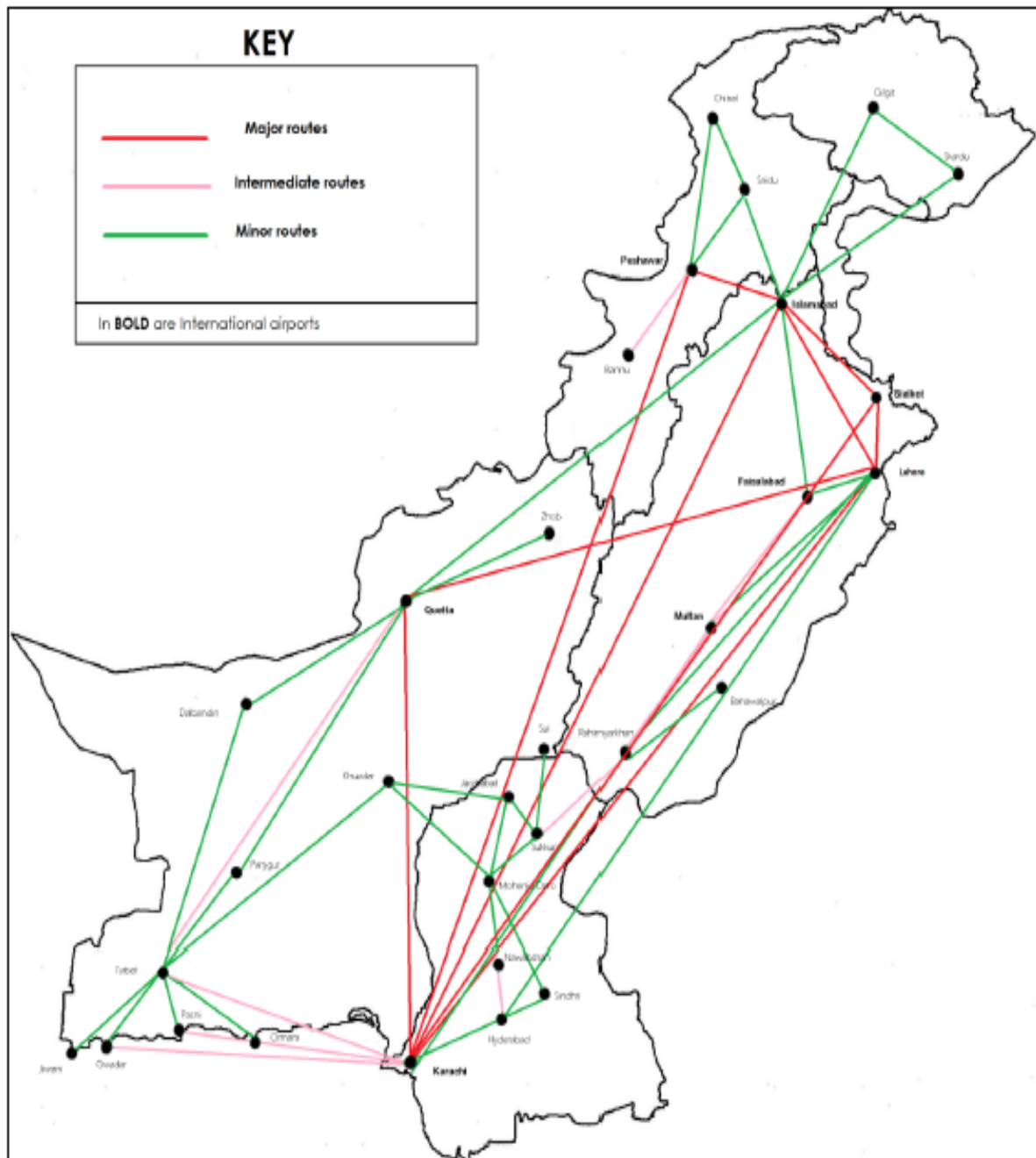
Benefits

- ✓ Chitral valley no longer cut off from the rest of Pakistan for 6 months per year.
- ✓ Access by road in winter when Lowari Pass closed by snow
- ✓ Shortens duration of journey to Peshawar by half (7 hours instead of 14 hours)
- ✓ Greater access to hospital/university/airports (in Peshawar and Islamabad)
- ✓ Greater access for trade
- ✓ Greater access for tourists/higher income from tourism
- ✓ Allows greater provision of services/food in winter
- ✓ Stimulates industrial development/employment

Problem

- ✓ Too many visitors (commercialisation of culture)
- ✓ Maintenance cost
- ✓ Easy movement for terrorists
- ✓ Increase in air pollution from exhaust fumes (which creates breathing difficulties)
- ✓ Loss of scenic beauty

AIR ROUTE OF PAKISTAN



DISTRIBUTION OF AIR ROUTES

- ✓ Most of the air routes follow the South-North direction, with less following East-West direction
- ✓ Most of the air routes run across the Indus Plains
- ✓ Karachi has the most flights in the country. Lahore, Faisalabad, Multan, Quetta and Peshawar are also major foci. All these are international airports along with the newly built airport at Sialkot
- ✓ Some minor routes branch from major airports, like Bannu is only accessible by air from Peshawar
- ✓ There are few flights in the North of country and also in West Balochistan, where low population and extremes of climate limit the number of passengers and flights

REASON FOR THIS DISTRIBUTION

- ✓ There are many flights from Karachi and other major cities because they have an international airport, not only do they serve domestic commuters but also international commuters
- ✓ The airports in bold are located in heavily populated areas where there is a greater percentage of people who are rich enough to afford air transport
- ✓ These cities have good transport infrastructure so they are easy to access and thus are preferred over other distant airports
- ✓ Areas like Karachi and Lahore etc are located in plain areas, for an international airport a long runway is necessary. Thus to accommodate large planes runway construction is impossible in steep sided hills of Northern Mountains
- ✓ Areas like Quetta, Peshawar, Lahore, Karachi and Islamabad are administrative capitals, where important offices of government are located, which need to be connected

REQUIREMENTS FOR AN AIRPORT

- ✓ Cheap, flat and abundant land is required for making the most important part of airport that is the runway
- ✓ A lot of capital is required for the construction of lounges, parking facilities, control tower etc
- ✓ Facilities like water supply, electricity, sanitation along with security are necessary
- ✓ Trained officers of customs and traffic controllers are also required
- ✓ Good transport infrastructure like road and rail network is vital for success

- ✓ Nearby population which can use the airport is also a pre-requisite. It's no use building an airport far away from population for example in a desert

Describe the problems in providing air transport in the northern areas of Pakistan.

- ✓ Mountainous area/rugged landscape
- ✓ Little level ground for runways
- ✓ Difficult landings/take-offs for pilots/restricted to small aircraft
- ✓ Frequent poor/bad weather for flying conditions
- ✓ Poor road access to airports
- ✓ Blocked telecommunications/radio
- ✓ Service unreliable causing flight cancellation

IMPORTANCE OF AIR TRANSPORT

- ✓ It is quick and cheap over long distances, it can serve other continents which is impossible by road and rail
- ✓ It is suitable for high value and light weight goods like computers both for import and export etc
- ✓ It can serve areas which are cut off to rail and road networks like Skardu
- ✓ It is mostly used by tourists coming to Pakistan and is also important for dispersing vital food and medical supplies during for example earthquakes etc
- ✓ Employment in airline industry
- ✓ More visitors/tourists (e.g. adventure tourists to northern areas/mountains)
- ✓ More business trips

RECENT DEVELOPMENTS AND FUTURE PLANS

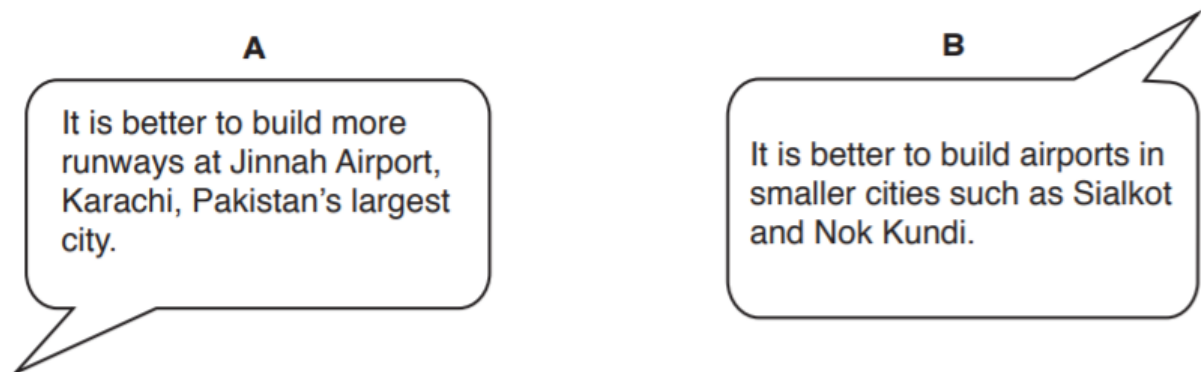
- ✓ The Sialkot Chamber of Commerce has completed the construction of Sialkot International Airport, thus making easier the export of high value and low weight goods, which were previously transported to Lahore airport 125km away. So cost of exports from the city has decreased. Also, the apron will be extended to accommodate more aircrafts with addition of construction of a jet bridge
- ✓ Multan International Airport will have an extension of runway from 9,000 to 11,000 feet, increasing the width of runway to 150 feet with 25 feet shoulders on each side E (Suitable for Boeing 747 aircraft). Other improvements will include expansion of terminal buildings. This will allow cargo flights to take place and thus suit the need of growing economy of Southern Punjab

- ✓ Gandhara Airport is being constructed to serve needs of Islamabad; it will become the biggest and most modern airport in Pakistan when it is completed in 2011

NAMES OF AIRPORTS

- ✓ Jinnah International Airport (Karachi)
- ✓ Islamabad International Airport
- ✓ Allama Iqbal International Airport (Lahore)
- ✓ Multan International Airport
- ✓ Peshawar International Airport
- ✓ Quetta International Airport
- ✓ Gwadar International Airport

Read the following two views about possibilities for air transport improvements in Pakistan:



Which view do you agree with more? Give reasons to support your answer. You should consider View A and View B in your answer. [6]

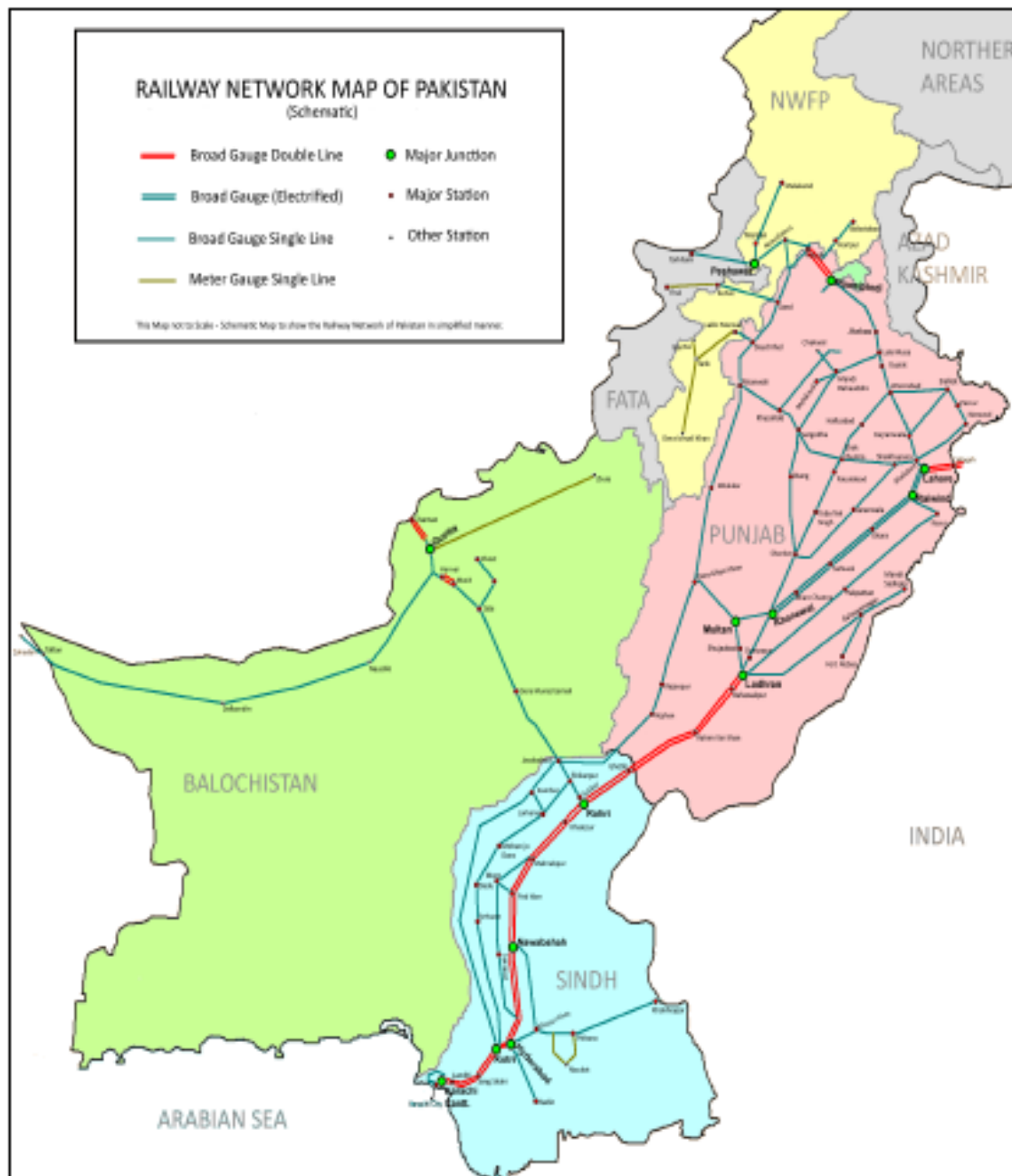
I agree with view A more.

Why?

- ✓ High population;
- ✓ Large demand for passenger travel
- ✓ Global airline companies more likely to fly to largest cities / provincial capitals
- ✓ Large international airport already present
- ✓ Cheaper to expand at current site than construct at a new site as air traffic control buildings already exist;
- ✓ Large industrial city;
- ✓ Good infrastructure facilities present.

- ✓ Nok Kundi very remote;
- ✓ Difficulties in building in Balochistan due to climate.

RAILWAY NETWORK OF PAKISTAN



DISTRIBUTION OF RAILWAYS

Sindh

- ✓ The rail network in Sindh starts from Karachi, where broad gauge double lines lead to Hyderabad, and then follow the bank of Indus into Northern Sindh towards Rohri
- ✓ A broad gauge single line branches from Hyderabad and heads onto Khokrapar and then into India. Also many broad gauge single lines connect small cities in Sindh which are west of Indus including Dadu and Larkana etc

Balochistan

- ✓ From Rohri in Sindh, a broad gauge single line branches and heads to Quetta in Balochistan. From Quetta a broad gauge double line leads to Afghanistan through Chaman, while another single line leads to Iran via Dalbandin and Taftan. Density of rail network is very low

Punjab

- ✓ From Rohri the double broad gauge continues till Lodhran
- ✓ From Khanewal to Raiwind and ultimately Lahore a broad gauge double line (electrified) runs
- ✓ From Lahore a broad gauge double line then runs through Wagah into India
- ✓ The pattern of railways across the most of Punjab is across the doabs and across the banks of Eastern tributaries of River Indus. Broad gauge single line are the most common

Khyber-Pakhtunkhwa

- ✓ From Rawalpindi a broad gauge double line operates till Attock, after which all major cities of Khyber-Pakhtunkhwa are served by broad gauge single line or narrow gauge single line. Density of rail network is low

IMPORTANCE OF RAIL TRANSPORT

- ✓ Rail transport is cheap over long distances inside a country
- ✓ It is also quick for long distances within a country
- ✓ It can carry the goods in bulk, which is impossible with road transport and to a limited extent with air
- ✓ It is suitable for low value, high weight cargo
- ✓ It is not affected much by traffic
- ✓ It is vital for success of dry ports and for deep sea port of Gwadar, unless Makran coast is provided with a modern rail link the prospects of Gwadar developing are bleak

LIMITATIONS OF RAIL TRANSPORT

- ✓ The maintenance of railway tracks is a very costly process, involving much labour and capital
- ✓ It is expensive to send small amounts of goods over a short distance
- ✓ Railways operate by a fixed schedule so rail transport isn't flexible nor is it door to door
- ✓ A lot of time is wasted during unloading and uploading of cargo, during which it can be either stolen or lost as different trains operate on different gauges
- ✓ The rail infrastructure in Pakistan is very old and unreliable prone to accidents which kill many hundreds of people annually

RECENT DEVELOPMENTS AND FUTURE PLANS

- ✓ It was announced in 2007 that a railway line between Gwadar and Quetta will be built and the Bostan-Zhob narrow-gauge railway line will be converted into broad gauge at a cost of US\$1.25 billion.
- ✓ Plans to increase train speeds, install more lengths of double track and to convert the country's railways to standard gauge are also currently under work.
- ✓ A feasibility study has been initiated to construct a rail link between Havelian (in Mansehra district Pakistan) and Kashgar (in China) via Khunjerab Pass. Thus giving China access to all year around warm waters of Indian Ocean
- ✓ A \$2 billion project for whole electrification of railways was launched in 2005 to continue till 2010

- ✓ A mass transit system is planned for Lahore to be completed by 2020; it will serve many hundreds and thousands of commuters daily. Also a mass transit system is planned for Karachi

To what extent is it possible to develop railways further in Pakistan? Support your answer by using examples you have studied. [6]

Possible

- ✓ Government/private funding so providing faster trains, e.g. Shalimar Express/launching new services/computerising ticket system
- ✓ Electrification
- ✓ Changing single track to dual (e.g. Khanewal to Lodhran)
- ✓ Foreign funding (e.g. Karakoram Express, Lahore to Karachi, launched 2002, mostly funded by China)

Not possible

- ✓ Hilly/difficult terrain (especially SW Balochistan/high mountains in N/NW)
- ✓ Much of network single line/track
- ✓ Lack of maintenance over a long period (worn out rails/sleepers)
- ✓ Inefficiently managed (overstaffed/operational inefficiencies / delays / corruption)
- ✓ Outdated
- ✓ Lack of funding

BENEFITS OF TRANSPORT NETWORK

- ✓ Quicker / cheaper long distance travel for passengers / goods;
- ✓ More developed public transport network means less need for car ownership;
- ✓ Employment
- ✓ Greater mobility of labour / university students;
- ✓ Greater ability to transport bulky / low value goods / freight / minerals;
- ✓ Rail development reduces number of cars / lorries on the roads and amount of air pollution
- ✓ Infrastructure improves

Explain how better transport routes could help to increase development in Balochistan.

- ✓ Industrialisation - bigger lorries, employment
- ✓ Urbanisation - better travel, less nomadism
- ✓ Faster travel for cars and lorries
- ✓ EPZ and dry port developed
- ✓ Better access to port
- ✓ Travel to Afghanistan or Iran via Quetta and passes
- ✓ Promote Tourism
- ✓ Better access to markets
- ✓ Living standards/quality of life gets better

To what extent could the improvement of road, rail and air transport improve the distribution of food supplies in Pakistan?

Possible

- ✓ air quick for perishable food
- ✓ rail slow but can be used for bulky goods
- ✓ road goes everywhere, door-to-door

Problems

- ✓ air expensive
- ✓ roads congested
- ✓ rail lack of maintenance, not door-to-door
- ✓ lack of funding, difficult topography, poor maintenance

PROBLEMS OF TRANSPORT NETWORK

- ✓ Rail not door-to-door
- ✓ Fumes from diesel trains cause air pollution / very polluting
- ✓ Noise pollution
- ✓ Railway tracks / motorways use up farmland;
- ✓ Railway tracks / motorways destroy natural habitat due to deforestation;
- ✓ Relocation of people / loss of homes / reduces value of property.

IMPORTANT PASSES

Pass to China

- ✓ Khunjerab Pass

Pass to Afghanistan from Peshawar

- ✓ Khyber Pass

Pass to Afghanistan from Baluchistan

- ✓ Khojak Pass

Pass to India

- ✓ GT Road (Grand Trunk Road)

Pass to Iran

- ✓ RCD Highway

DRY PORTS

- ✓ They are inland ports, which are basically terminals inland and are directly connected to ports by either rail or road networks.
- ✓ They can be export or import orientated.
- ✓ For example the Sialkot dry port is export orientated meaning that it is organized in such a manner that in which it is much easier to export goods as compared to importing goods.

At present, there are some dry ports running under the management of Pakistan Railways:

- ✓ Lahore Dry Port Established in 1973
- ✓ Quetta Dry Port Established in 1984
- ✓ Peshawar Dry Port Established in 1986
- ✓ Multan Dry Port Established in 1988
- ✓ Rawalpindi Dry Port Established in 1990

In addition to the above, there are Dry Ports established and running under the management of private sector

- ✓ Sialkot Dry Port Established in 1986. Busiest dry port in Pakistan. First private dry port in Asia
- ✓ Faisalabad Dry Port Established in 1994
- ✓ Pak-China Sust Dry Port
- ✓ NLC Dry Port at Thokar Niaz Beg Lahore
- ✓ NLC Dry Port at Quetta

Features of a dry port are:

- ✓ Facilities for storage, preservation and consolidation of goods
- ✓ Refrigeration facilities
- ✓ Loading area with lifts and cranes etc
- ✓ Custom clearance services
- ✓ Connection with road and railway infrastructure

IMPORTANCE OF DRY PORTS

- ✓ Dry ports have custom clearance services, which mean that goods are cleared before they reach port, thus saving hassle at port. Time is saved this way and goods can be exported easily and on schedule.

- ✓ Storage space is saved at the port area as goods are only sent from the dry port only when they are needed by the schedule
- ✓ Furthermore, containerization means that there is less of a chance to damage to goods or that they may get stolen, as previously goods had to be sent to port where they would be then put in containers
- ✓ Taxes can be collected at the dry port. Also, less storage space is required at the sea port, thus freeing up land for extension of port
- ✓ Refrigeration facilities are present so perishable items can be stored and need not to be delivered directly to the port immediately.
- ✓ Transport charges are reduced as now goods are transported in bulk and this lowers the cost to importer/exporter.
- ✓ Producers inland can easily export their products (at higher price than what they would have sold to domestic consumers), this incentivizes them to produce more and improve the quality of their goods (to attract more foreign customers)

Explain how dry ports have increased trade in Pakistan.

- ✓ Increases foreign trade / more exports / more imports
- ✓ Better access to remote areas / areas away from Karachi
- ✓ Saves time / expense of providing own transport to coast
- ✓ Less congestion / relieves burden at Karachi / sea ports
- ✓ Speeds up / more efficient paperwork / customs checks
- ✓ Encourages investors to invest
- ✓ More efficient loading / containerisation

COMMUNICATION

- ✓ Successful communication occurs when the sender sends the correct message to the receiver, who understands the message and if necessary gives a reply acknowledging that he has understood the message and if necessary acted upon it
- ✓ Pakistan has around a million broadband users and around 100 million mobile subscribers. The growth rate of telecom industry is one of the highest in the world

MEANS OF COMMUNICATION

- ✓ Internet/Email
- ✓ Telephone
- ✓ Mobile
- ✓ Fax
- ✓ Radio
- ✓ Video Conferencing

IMPORTANCE IN BUYING AND SELLING

- ✓ Communication helps in locating the supplier, as the customer can search and then order the goods online etc or on the phone. This is done within a country and also internationally. This is much quicker and cheaper than going to the offices of the producer etc
- ✓ The producer can advertise his goods online etc. He can also show his customers the progress of their orders. Furthermore, the mode of transport and the means of payment can both be decided

IMPORTANCE IN DEVELOPING INDUSTRIES

- ✓ Machinery can be searched over the internet, prices and specifications of different models can be compared. Pictures and videos of demonstration models can also be seen. Reviews can also be read, therefore, the best choice can be made
- ✓ Advertising can increase sales and market share, thus increasing a company's profit and eventually freeing up capital for investment in the business
- ✓ People for important posts can be hired through online advertisements; this means that the best people from a wide pool are selected for the job

- ✓ Foreign investors can read news of promising new developments in other countries. For example news of Thar Coal deposits have generated excitement in mining circles. This interest can fuel the proceeding contract bidding process and help make sure that the best bid is selected
- ✓ Bank policies and interest rates can be determined online or on the phone. Online payments can also be made quickly, so there is no need to go to banks (saves time)
- ✓ Furthermore, training of employees can be done by showing them video clips etc. Research papers into new techniques can be read and the techniques then implemented in the workplace
- ✓ Video conferencing can be done between managers in different cities and countries, thus a decision which is beneficial to all of the branches of a company can be implemented

IMPORTANCE IN EDUCATION

- ✓ Communication can play a vital role in the very important sector of education. Pakistan has the largest Wimax network in the world today
- ✓ This can be used to broadcast television lectures to children in schools etc. Lectures can be recorded and put online. They can be listened to again and again, rewinded and moved forward. People can now learn at their own will. Videos can be made of complex models and also put online. All of this means that one teacher can do the work of a million teachers to a certain extent

IMPORTANCE IN ALLEVIATING RURAL POVERTY

- ✓ Use of telecommunication is probably the most under estimated way of alleviating rural poverty in Pakistan. As demonstrated above if the field of education can be handled via telecommunications then this can help in controlling population growth. This is so because rural communities will be able to realize the benefits of smaller families
- ✓ Also telecommunications help in increasing agricultural productivity; farmers can be demonstrated new methods of planting, growing and harvesting crops. Also, weather patterns can be shown to farmers (rain forecast, chance of pest attack etc) so they can plan their processes. This will result in increasing yields and will lead to increased profits, so more money will be available for investment in fertilizers, insecticides etc